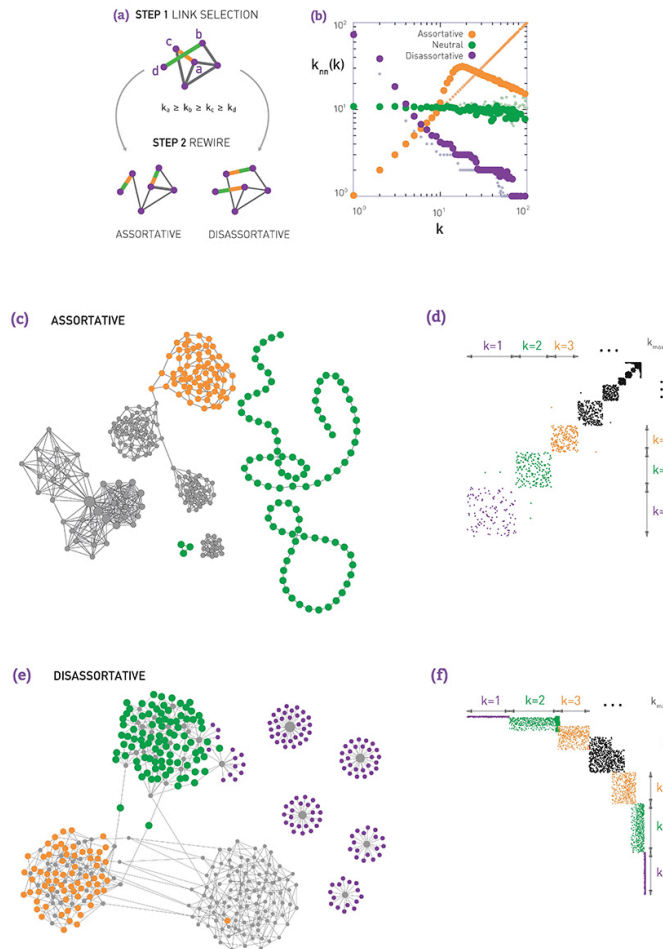


L12: Generating Networks with Degree Correlations

Another important network property is the presence of correlations in the degrees of adjacent neighbors. As we have seen in Lesson-3, in **assortative** networks nodes tend to connect to nodes with similar degree, while in **disassortative** networks the opposite happens. How can we create networks that exhibit assortative or disassortative behavior?

Suppose that we are given a network (it could be a network that is generated from another model). We can apply the procedure shown in the following visualization:



- Select two random links. Let us call the corresponding four end-points (stubs) as (a,b,c,d), ordered based on their degree so that $k_a \geq k_b \geq k_c \geq k_d$.
- If we want to create an assortative network, we can rewire the two links so that the two higher-degree nodes connect, adding the edge (a,b) if that edge does not already exist. Also, we connect the two lower-degree nodes, adding the edge (c,d) if that edge does not already exist.
- If we want to create a disassortative network, we do the opposite: we connect the highest-degree node (node a) with the lowest-degree node (node d), and node b with node c.

The previous process is applied iteratively until we cannot find pair of edges that can be rewired.

The visualization (b) shows what happens when we apply the previous process on a network generated with the preferential attachment model (with $N=1000$ nodes, and $L=2500$ edges). The plot shows the average neighbor node degree $k_{nn}(k)$ as a function of the degree k (see Lesson-3 if you do not recall such plots). Without applying the previous algorithm, the original network is “neutral”, without any significant degree correlations.

When we apply the previous algorithm to create a disassortative network, the degree correlation between a node and its neighbors become strongly negative (purple points).

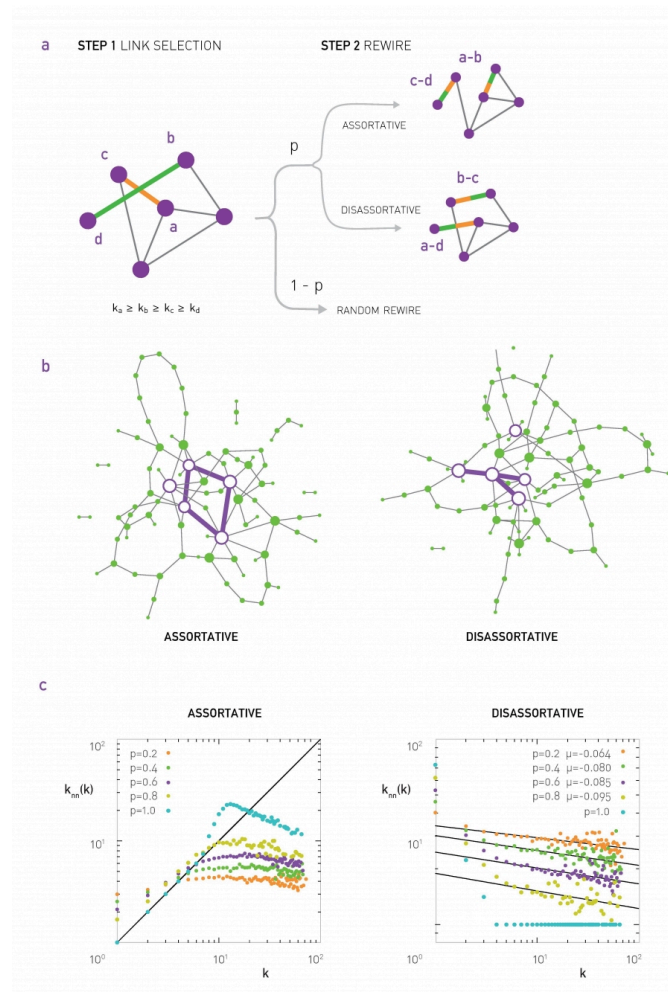
On the other hand, when we apply the previous algorithm to create an assortative network, the degree correlations become positive at least up to a certain degree (the reason that the degree correlations become negative for higher degrees is related to the constraint that two nodes can connect with only a single link – the high-degree nodes are few and to maintain a positive degree correlation for high-degree nodes would require several links between the same pair of nodes).

Visualizations (c and d) show an example of an assortative network generated by the previous algorithm and the corresponding adjacency matrix. Note that nodes of degree-1 connect to other nodes of degree-1, while nodes of higher degree connect to other nodes of similarly high degree.

Similarly, visualizations (e and f) show an example of a disassortative network generated by the previous algorithm, and the corresponding adjacency matrix. In this case, nodes of degree-1 connect to the highest degree nodes, while there is also a large number of intermediate-degree nodes that connect to other intermediate-degree nodes.

The previous approach produces maximal assortativity or disassortativity because it keeps rewiring edges to maximize the extent of the degree correlations. Another approach is to apply the previous process *probabilistically*: with some probability p

we perform the previous rewiring step, and with the complementary probability $1-p$ we leave the two randomly selected edges unchanged. The higher the value of p , the stronger the degree correlations will be. The following visualization illustrates this process, shows two networks generated in this manner, and it gives average neighbor node degree plots for various values of p .



Food For Thought

Explain mathematically the reason we get negative degree correlations even in assortative networks, when we require that two nodes can be connected by at most one edge. Hint: In the configuration model, what is the expected number of edges between two nodes with degree k_i and k_j ?